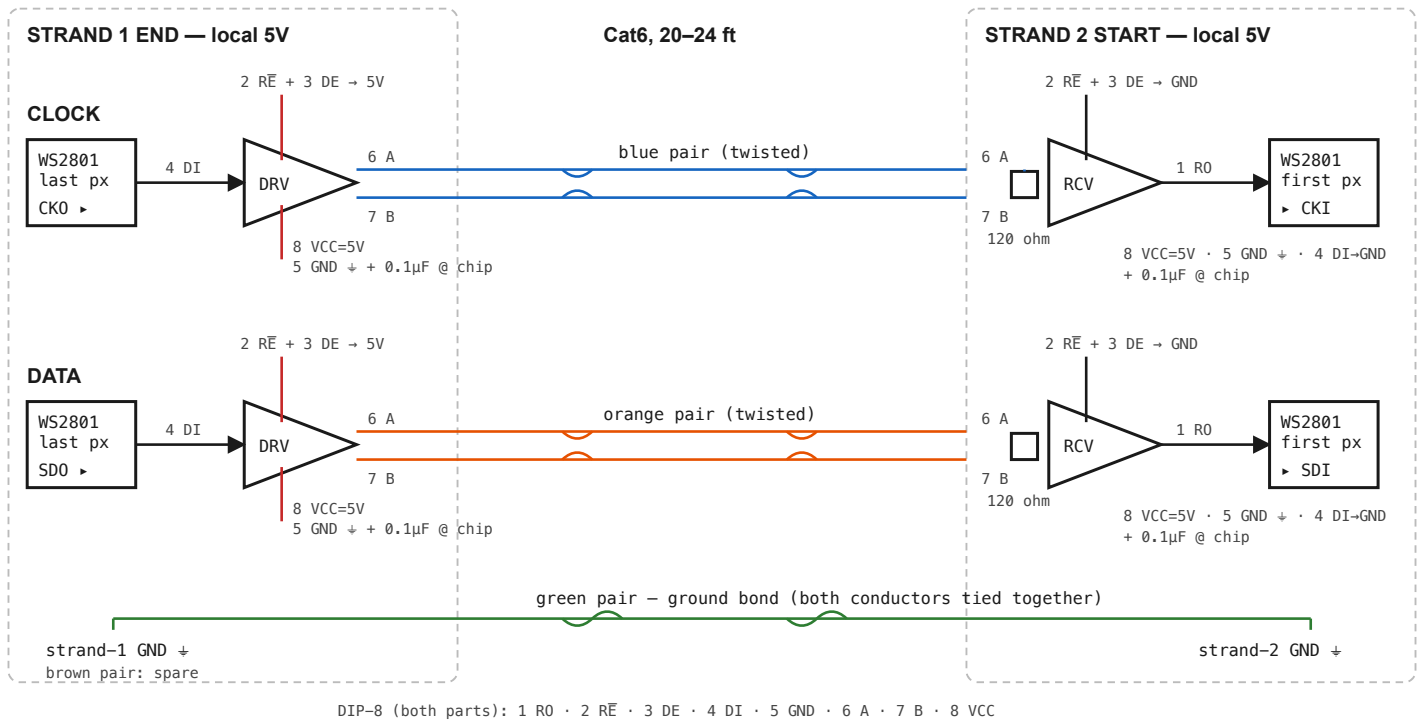


RS-485 Differential Extension — WS2801 Inter-Strand Bridge

Bridges the ~24 ft gap between LED strand 1 (end) and strand 2 (start) of one 800-pixel WS2801 chain, carrying the 5 V CKO/SDO clock and data differentially over one Cat6 run. Simplex, one direction, no direction switching. · Lightbringer project · 2026-08

Enable mnemonic — DRIVER end (strand 1): pins 2+3 tied HIGH (5 V). RECEIVER end (strand 2): pins 2+3 tied LOW (GND).

Schematic



Four chips total: 2 drivers at the strand-1 end, 2 receivers at the strand-2 start — one each for CLOCK and DATA. ST485BN and MAX485CPA are interchangeable here (identical pinout, 5 V, 2.5 Mbps); use four of the same part if stock allows, otherwise keep each signal path homogeneous (both CLOCK chips one type, both DATA chips the other). Each end is powered from the **local 5 V** injection already present at its strand — no logic power crosses the gap.

Driver end (strand 1) — per chip

Pin	Connect to
4 DI	WS2801 output: CKO (clock chip) / SDO (data chip), leads in inches
3 DE	5 V — driver always enabled
2 RE	5 V — unused receiver disabled
1 RO	no connect
8 VCC	local 5 V injection
5 GND	local GND

Receiver end (strand 2) — per chip

Pin	Connect to
6 A / 7 B	matching Cat6 pair + 120 Ω across A–B (this end only)
2 RE	GND — receiver always enabled
3 DE	GND — unused driver disabled
4 DI	GND (tie off)
1 RO	WS2801 input: CKI (clock chip) / SDI (data chip), leads in inches

6 A / 7 B	one Cat6 pair, out to far end	8 VCC	local 5 V injection
—	0.1 μ F pin 8→5, at the chip	5 GND	local GND
		—	0.1 μ F pin 8→5, at the chip

Cat6 pair assignment

Pair	Carries	Rule
Blue	CLOCK A/B	A and B of one signal must be the two wires of the same twisted pair — never split a signal across pairs, never share a pair between clock and data
Orange	DATA A/B	
Green	Ground bond	both conductors tied together, connected to GND at both ends — required, not optional
Brown	spare	reserve for a second ground or future signal

Never run LED power over the Cat6. The gap carries the clock pair, data pair, and ground bond — nothing else. No series resistors on the short chip↔WS2801 hops (they are inches, not transmission lines); the only termination in this stage is the 120 Ω at each receiver.

Parts list (one bridge)

Qty	Part	Note
4	ST485BN or MAX485CPA, DIP-8	2 drivers + 2 receivers; DIP-8 sockets recommended
2	120 Ω , ¼ W	termination, receiver end only, one per pair
4	0.1 μ F ceramic ("104")	one per chip, pins 8–5, close to the chip
1	Cat6/6A, 20–24 ft	solid bare copper, unshielded is fine
2	small perfboard + terminals	one per end

Bring-up sequence

- Wire and power both ends from their local 5 V. With the controller idle, a scope across A–B at the receiver shows a clean differential idle (the driver is always on — the line is never floating).
- Run at the chain's standard clock — currently **1.5 MHz** (`hw.led.ins[0].freq: 1500`, validated on the installed array). Push full white plus a fast effect; watch **strand-2** pixels for sparkle or wrong colours.
- If strand 2 shows corruption, drop to **1 MHz** (`freq: 1000`) — a runtime config change, no reflash. 800 px at 1 MHz still refreshes ~52 Hz, above WLED's 42 fps target: no visible cost.
- Pass criterion: strand 1 and strand 2 pixel-identical side by side under motion.